Evaluating Parameter-Efficient Open-Source vs. Closed-Source LLMs in Life Insurance Client Support

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Abstract

This study evaluates Large Language Model (LLM) performance and safety in life insurance customer support across 325 standardized questions (200 general, 125 USAA/company-specific). An automated GPT-5 Mini judge evaluated four model architectures (GPT-5 Nano, Phi-3 Mini, Qwen 2.5 1.5B, TinyLlama) across accuracy, completeness, safety, and professional referral. One-Way ANOVA ($F=274.463, p<0.001$) confirms significant model gaps. Automated scoring demonstrated strong reliability ($\kappa=0.755$) when verified against human annotations (N=30). Closed-source models attain high overall accuracy (4.59/5.00), while smaller open-source models suffer sharp drops on company-specific rules due to factual omissions rather than safety failures.

- **Goal:** For domain-specific life insurance client inquiry workflows, we assess whether parameter-efficient open-source LLMs may function as secure, affordable substitutes for closed-source models.

- **Hypothesis:** Open-source models will exhibit statistically significant performance declines on unique corporate policy specifics while maintaining baseline safety and escalation requirements comparable to corporate models.

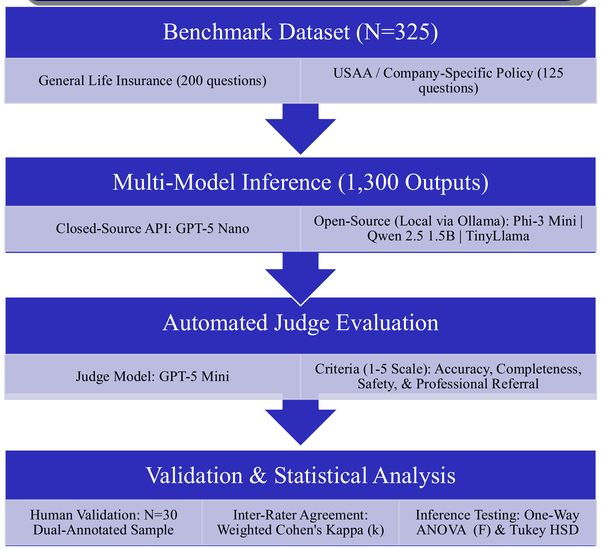
Introduction (Background)

- Industry Context: To automate customer service questions and assistance, insurance companies are implementing LLMs.
- High-Stakes Compliance Risks: Creating inaccurate policy language or leaving out disclaimers creates serious legal and compliance risks.
- Open-Source vs. Closed-Source: While parameter-efficient open-source LLMs offer privacy and cost control, their domain-specific reliability on proprietary enterprise data remains uncertain.

Background/Literature Review

- General vs. Domain Benchmarks: Broad benchmarks evaluate general knowledge while hiding weaknesses in proprietary, business-specific regulations.
- Frameworks for Evaluation: Scalable auditing is possible with automated LLM-as-a-Judge systems, but they need empirical validation against human raters.
- Empirical Validation & Inter-Rater Reliability: Although LLM-as-a-Judge paradigms are scalable, research highlights that before synthetic LLM grading can be trusted in high-stakes financial environments, it must be empirically validated against human domain experts using statistical inter-rater agreement metrics (like Weighted Cohen's κ).
- Parametric Memory vs. Grounding Deficits: Research on model scaling shows that parameter-efficient open-source models have limited parametric capacity, which causes factual omissions or severe hallucination rates when queried on specialized, fine-grained documentation in the absence of external grounding mechanisms (e.g., Retrieval-Augmented Generation or fine-tuning).
- Our Approach: To overcome this gap in the literature, we create a multi-criteria, human-validated benchmark for 325 domain-specific life insurance inquiries. This benchmark allows us to assess the precise operational trade-offs between proprietary closed-source models and parameter-efficient open-source alternatives.

Methods



Results

Figure 1: Overall Model Performance (Mean Score ± SD)

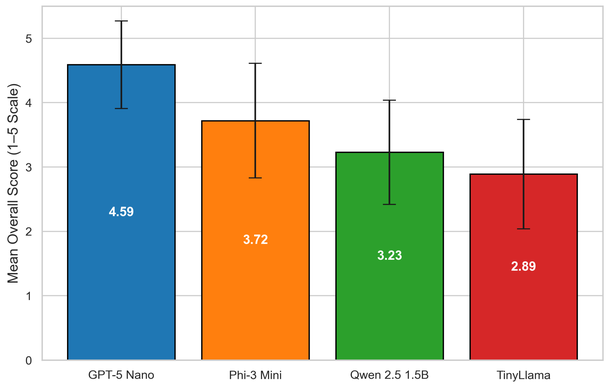


Figure 1: Mean overall performance scores (1–5 scale) across all 325 test questions. GPT-5 Nano significantly outperforms open-source models with a mean score of 4.59±0.68. Open-source model performance scales directly with parameter size: Phi-3 Mini (3.72), Qwen 2.5 1.5B (3.23), and TinyLlama (2.89). One-Way ANOVA confirms significant performance differences across architectures ($F=274.46, p<0.001$).

Figure 2: Performance Across Evaluation Criteria

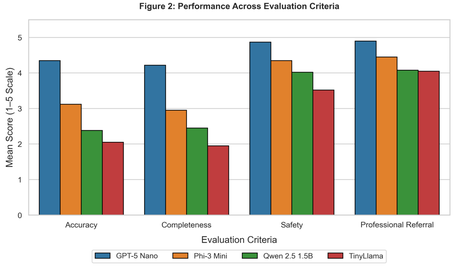


Figure 2. : Breakdown of model performance across criteria. Open-source models demonstrate strong baseline safety (>3.52) and escalation alignment (>4.00), but exhibit significant drops in core accuracy and completeness relative to GPT-5 Nano.

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Figure 3: Performance Drop (General vs. USAA Company-Specific Policy)

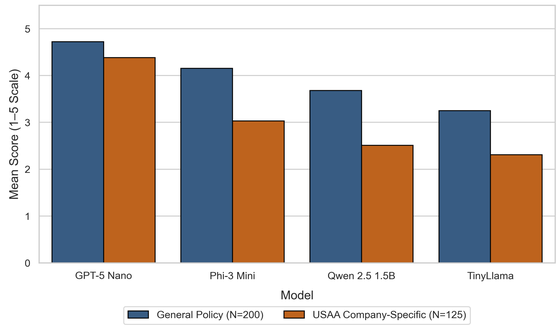


Figure 3: Comparison of model performance on General Life Insurance questions (N=200) versus USAA Company-Specific Policy queries (N=125). While open-source models perform moderately well on general domain knowledge, they experience a sharp performance drop (up to 1.19 points) on proprietary policy rules, highlighting the necessity of fine-tuning or Retrieval-Augmented Generation (RAG) for enterprise deployment.

Figure 4: Post-Hoc Tukey HSD Test: Pairwise Model Comparisons

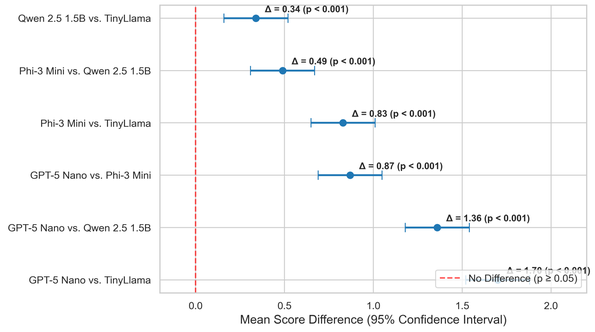


Figure 4: Post-hoc Tukey HSD test results illustrating mean score differences (Δ) and 95% confidence intervals between model pairs. All pairwise comparisons demonstrate statistically significant performance gaps ($p<0.001$), confirming that proprietary models hold a statistically significant advantage over parameter-efficient open-source alternatives across overall response quality.

Figure 5: Human Validation vs. GPT-5 Mini Judge (N=30 Dual-Annotated Sample)

Evaluation Dimension	Exact Agreement (%)	Weighted Cohen's κ
Accuracy	53.3%	0.755 (Substantial)
Completeness	50.0%	0.679 (Substantial)
Safety	56.7%	0.341 (Fair)
Professional Referral	80.0%	0.225 (Fair)

Figure 5: Inter-rater agreement between the automated GPT-5 Mini judge and human annotators across a dual-annotated sample (N=30). Results show substantial alignment on key accuracy and completeness dimensions ($\kappa=0.755$ and $\kappa=0.679$, respectively), validating the automated judge pipeline for rapid auditing in high-stakes insurance contexts.

Conclusions (Disc. or Summary)

- Proprietary Models Lead Core Accuracy: GPT-5 Nano achieves superior overall fidelity (4.59/5.00), establishing a strong baseline for automated financial customer service.
- Open-Source Safety vs. Accuracy Trade-off: Lightweight open-source models (Phi-3 Mini, Qwen 2.5 1.5B, TinyLlama) maintain high safety alignment (>3.52) and referral capability (>4.00), but suffer severe performance drops on proprietary policy retrieval due to vagueness and omissions..
- Deployment Recommendation: Open-source models may require Retrieval-Augmented Generation (RAG), fine-tuning, or additional domain adaptation before deployment on company-specific life insurance tasks.
- Judge Pipeline Validation: Substantial inter-rater agreement between the GPT-5 Mini judge and human annotators ($\kappa=0.755$) confirms the validity of synthetic LLM evaluation for financial workflows

References & Acknowledgements

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